

Cultural Understanding in Large Language Models: A Comparative Study with Human Judgments

Elsayed Issa
Purdue University
West Lafayette, IN, USA
esissa@purdue.edu

Kamilia Rahmouni
Virginia Commonwealth University
Richmond, VA, USA
rahmounik@vcu.edu

Abstract

This paper evaluates whether large language models (LLMs) can demonstrate cultural competence in Egyptian Arabic—not just factual knowledge about culture, but the ability to choose culturally appropriate idiomatic and figurative expressions in everyday social situations. 67 Egyptian participants completed a multiple-choice questionnaire of 90 everyday scenarios to provide human gold labels. Using the same scenarios, each pairing a culturally appropriate Egyptian idiom against a plausible distractor, we evaluated 11 open-source and 3 proprietary models under five prompts and two answer orderings. The results indicated that humans performed near ceiling (98.2%), while LLMs averaged 68.2%. The results also showed that a single accuracy figure conflates three separable behaviors: cultural *competence*, an answer-*position* heuristic, and *abstention*. These behaviors are distributed so unevenly that the model ranking reorders depending on which one is measured. One model’s apparent mid-tier accuracy is almost entirely a “pick the first option” heuristic (a 0.75 normal–reversed gap), and another falls from 51.8% to below chance once refusals to answer are scored as errors. We argue that cultural-competence evaluation should report this decomposition rather than a headline accuracy.

1 Introduction

Large language models (LLMs) have made remarkable advances in natural language processing, from text generation and translation to question answering (Brown et al., 2020; Raffel et al., 2020). Yet these advances are contingent on vast training corpora dominated by high-resource languages (Baack, 2024). As a result, most LLMs struggle to capture the breadth of global cultural contexts. Studies show that LLMs frequently fall short on fairness and cultural diversity, resulting in biased outputs that privilege some groups while disadvan-

taging, and potentially harming, others (Sambasivan et al., 2021; Hofmann et al., 2024; Gallegos et al., 2024).

Culture is a complex, multidimensional concept that describes values, beliefs, norms, and ideas that comprise a way of life (Murdock, 1940). Building on this, cultural competence can be defined as the alignment of behaviors, attitudes, policies, and structures within a system, agency, or professional group that enables effective work in cross-cultural contexts (Cross et al., 1989). In language-related domains, this notion extends to the ability to interpret and produce utterances in ways that align with community-specific cultural expectations, including sensitivity to idiomatic, figurative, and context-dependent meaning (Kramsch, 2014).

Much existing work evaluates the *declarative* cultural knowledge of LLMs rather than situated language use (Hovy and Yang, 2021; Adilazuarda et al., 2024). This gap is acute for under-represented dialects such as Arabic varieties, where meaning is often conveyed implicitly through idioms (Naous et al., 2024; Attia et al., 2025). A natural response is to construct scenarios, score models against native speakers, and report the gap. We built such dataset for Egyptian Arabic, and in the process discovered that the headline metric it produces is not measuring what it appears to.

This paper asks two questions. First, do LLMs trail native speakers on everyday Egyptian-Arabic use? They do, but a verdict alone is uninformative about *why*. So we also ask what a cultural-competence accuracy score is actually made of. On a controlled forced-choice task, a single accuracy number conflates three distinct behaviors: pragmatic *competence*, a preference for an answer *position*, and *abstention*, where a model declines to answer. Because our design presents every item in both answer orderings and records refusals, we can separate these components, and once separated they prove more informative than the aggregate.

2 Related work

There has been considerable work that systematically benchmark the cultural dimensions of LLMs. For example, Wang et al. (2024b) operationalized culture through established cultural dimensions and probed models with questions spanning multiple domains (e.g., family, work, social life) to see which cultural profiles their responses most closely match. CulturalBench (Chiu et al., 2025), for instance, assembled human-verified questions that cover 45 regions and 17 topics ranging from food and etiquette to holidays and local norms. Findings revealed gaps between model and human performance, especially for underrepresented regions. Other studies proposed region- or language-specific benchmarks—such as datasets targeting Saudi dialects (Ayash et al., 2025) or Chinese–Korean cultural knowledge (Lee et al., 2025)—to test whether models can adapt to local practices rather than defaulting to Anglophone or globalized norms.

Several studies used figurative and culturally grounded language as a proxy for local knowledge and cultural competence. Studies on multilingual proverbs (e.g., MAPS (Cecilia Liu et al., 2024)) demonstrated that LLMs often recognize proverb surface forms but struggle to reason with their figurative meanings, especially when expressions are translated or are less represented. Similar findings were salient in work on proverb translation and on catchphrases, memes, and proverbs in Russian (Gromenko et al., 2025; Wang et al., 2025).

Furthermore, Liu et al. (2022) proposed Fig-QA, a benchmark that tests whether LLMs can correctly map figurative expressions to their intended non-literal meanings. Chakrabarty et al. (2022a) presented a benchmark of fictional narratives containing idioms and similes, where models must select or generate a plausible continuation that reflects the figurative meaning. Similarly, Chakrabarty et al. (2022b) developed FLUTE, an NLI-style figurative language benchmark that pairs entailment/contradiction labels with free-form textual explanations across sarcasm, simile, metaphor, and idioms. The PUB benchmark (Sravanthi et al., 2024) is a multiple-choice dataset covering four pragmatics phenomena designed to systematically evaluate LLMs’ ability to infer meaning beyond literal semantics. Moreover, (Azime et al., 2025) introduced ProverbEval, a culture-grounded benchmark that uses proverbs to test LLMs in low-resource languages, showing how proverb interpretation relies

on language- and culture-specific meaning beyond surface-level translation.

With respect to Arabic, several studies examined how LLMs handle Arabic dialects in culturally grounded contexts. Ayash et al. (2025) introduced SaudiCulture, a benchmark evaluating LLMs’ understanding of Saudi culture across five geographical regions and multiple cultural domains including food, clothing, and traditions. Their evaluation of six models revealed significant performance declines on region-specific questions, with accuracy ranging from 16% to 66% depending on the model and region. Similarly, Yakhni and Chehab (2025) examined culturally-aware translation in Lebanese Arabic, identifying critical limitations in existing parallel datasets, particularly their non-native nature and absence of cultural context. Their evaluation demonstrated that while LLMs and neural MT systems perform comparably on non-native datasets, LLMs exhibit superior capability in preserving culture when processing Lebanese content.

Moreover, Naous et al. (2024) examined cultural bias in Arabic through contrastive everyday scenarios, introducing CAMEL, a resource contrasting Arab and Western cultural entities across naturally occurring prompts. Their evaluation showed that both multilingual and Arabic monolingual models systematically favor Western cultural associations and struggle to adapt to Arab cultural contexts. In addition, AlKhamissi et al. (2024) examined cultural alignment in Arabic and English by simulating sociological surveys conducted in Egypt and the US, comparing LLM responses to human survey data. Their results showed that cultural alignment improves when models are prompted in the dominant language of culture and when pre-training data better reflects that culture, while misalignment is more pronounced for culturally sensitive topics. For Arabic culture datasets, Sadallah et al. (2025) introduced ArabCulture, a culturally grounded commonsense reasoning benchmark across 13 Arabic cultures that included idioms as one of the categories as well as the Jawaher dataset (Magdy et al., 2025) that includes a large-scale collection of Arabic proverbs across 20 dialects.

Across the discussed research, culture in LLMs were assessed through benchmarks, figurative language tasks, translation performance, and region-specific datasets, revealing persistent Western-centric biases and uneven handling of cultural use. These limitations are particularly evident in Arabic, which remains underrepresented in NLP research

and continues to exhibit gaps between human and model interpretations of culture and idioms (Naous et al., 2024). Moreover, few studies have targeted deep behavioral adaptation in LLMs, and there is no established way to measure depth or progress toward *cultural competence* in LLMs (Liu et al., 2025). In response, the present study evaluates whether LLMs can demonstrate cultural competence in Egyptian Arabic—not just factual cultural knowledge, but the ability to choose culturally appropriate idiomatic and figurative expressions in everyday social situations, as humans do.

Additionally, our framing draws on a related literature on *evaluation artifacts*. LLMs are sensitive to option order in multiple-choice questions (Pezeshkpour and Hruschka, 2024; Zheng et al., 2023; Anghel et al., 2025) and to superficial prompt formatting (Sclar et al., 2023; Ceron et al., 2024), and first-token probabilities can diverge from generated answers (Wang et al., 2024a). Refusal and over-cautious behavior are themselves context-dependent (Cheng et al., 2026; Tao et al., 2024; Yong et al., 2025; Chae et al., 2025), and larger models can become *less* reliable on easy items (Zhou et al., 2024). We connect these threads to the cultural evaluation we conduct in this study.

3 Methods

3.1 Scenario Construction

To test Egyptian cultural and idiomatic phrases in both humans and LLMs, the first author, an Egyptian native speaker, constructed 90 short, everyday scenarios that depict familiar social situations (for example, unfair hiring or nepotism as Figure 1 shows). These scenarios were reviewed and validated by an Egyptian cultural expert as part of the Institutional Review Board (IRB) process. For each scenario, two candidate expressions, typically idioms or fixed phrases, were selected and only one of which is culturally appropriate (Table 6).

Domain	Count
Life wisdom and general advice	22
Everyday states and reactions	20
Character and reputation	18
Conflict and confrontation	10
Money and wealth	6
Marriage and gender relations	4
Work and institutions	3
Fate, luck and the supernatural	3
Family and kinship	2
Hospitality and social visits	2

Table 1: Distribution of scenarios by cultural domain

Table 1 reports the distribution of the 90 scenarios across social domains reflecting the breadth of situations in which culturally appropriate expressions arise. Each scenario was assigned to a single social domain following a taxonomy developed by the authors.

3.2 Human Data Collection

67 Egyptian speakers were instructed to read each scenario and choose the option (1 or 2) that best describes the situation via a Qualtrics survey. Participants were aged 20–44 (42 male, 25 female) and all held a higher-education degree. The survey was distributed in Alexandria, Beheira, and Cairo. This distribution bears directly on validity. If demographic variation within our pool materially affected judgments of idiomatic appropriateness, we would expect measurable disagreement across participants. Instead, agreement with the intended labels is near-unanimous (98.2%). This convergence indicates that the idioms tested are broadly shared across the sampled population.

3.3 Model Prompting

To evaluate LLMs on the same task, each scenario–option pair was converted into five prompt templates as Table 2 shows.

Prompts
1 Which phrase is the correct response to this scenario? 1, 2. Respond with 1 or 2 as your answer. Direct Simple Question
2 Which phrase is more acceptable as a response to this scenario? 1, 2. Respond with 1 or 2 as your answer. Acceptability Judgment
3 Which phrase are you, as a large language model, more likely to produce as a response to this scenario? 1, 2. Respond with 1 or 2 as your answer. LLM Production Choice
4 After reading this scenario, which phrase are you more likely to produce as a response? 1, 2. Respond with 1 or 2 as your answer. Production Choice
5 Which phrase is the correct response to this scenario? 1, 2. Respond with 1 or 2 as your answer. Justify your answer. Direct Simple Question with Justification

Table 2: Prompt templates used in the evaluation and their associated categories.

Prompt 1 is the direct simple question that mirrors the human survey format. Prompt 2 is adopted from questions in linguistic acceptability judgments, which can reveal implicit linguistic knowledge (Chomsky, 2002; Sprouse, 2011). Prompts 3 and 4 use production choice in general or as an LLM (Song et al., 2025). We used reversed prompts, asking the model to select one option

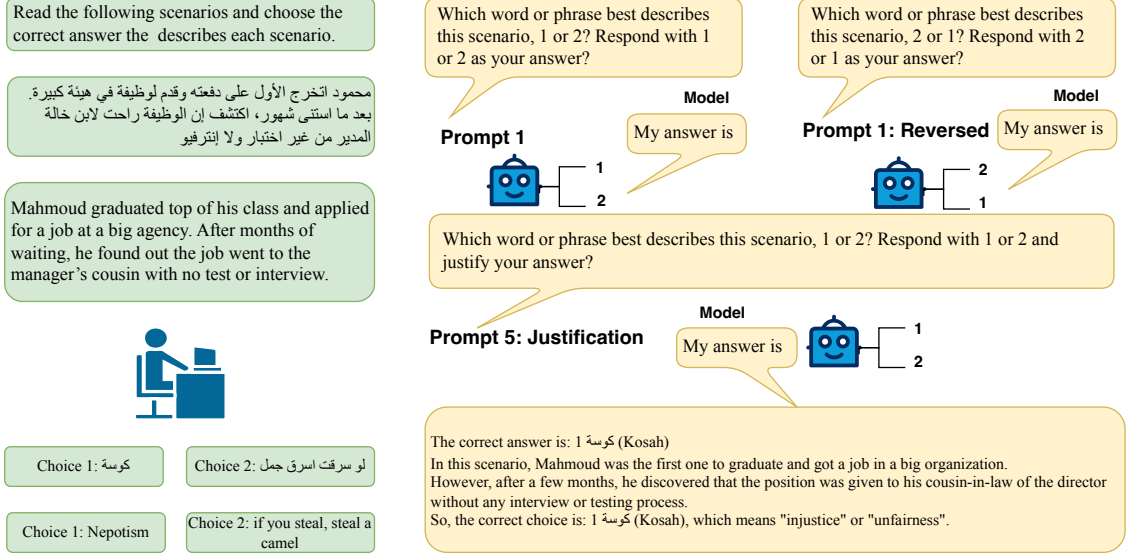


Figure 1: The scenario-based task and prompting protocol. Each scenario is presented with two phrases. Models answer under three conditions: (1) direct forced choice, (2) reversed label–number mapping to probe option–order bias, and (3) forced choice with brief justification. Human judgments are collected via a Qualtrics survey.

while varying the order of the choices to control for position biases. We also observed that some models were more likely to select the appropriate answer when asked to provide an explanation, so we introduced a prompt that requests a justification for the chosen option. Following (Wang et al., 2024a), we appended the string *My answer is* to the end of each prompt to mitigate issues associated with relying solely on first-token probabilities.

We prompted 11 open source LLMs with different sizes and 3 proprietary models (GPT 5.2, Claude Sonnet 4.5, and Gemini 3 Flash). These open source models included Qwen3 8b and 14b (Yang et al., 2025), Llama3 3b and 8b (Touvron et al., 2023), Gemma 4b and 12b (Team et al., 2025), DeepSeek 8b and 14b (Liu et al., 2024), AceGPT 7b and 13b (Huang et al., 2024), and Command r7b (Alnumay et al., 2025). This yields $14 \times 90 \times 9 = 11,340$ model responses, alongside $67 \times 90 = 6,030$ human judgments. Exact checkpoints and APIs, access dates, and system-prompt usage are reported in Appendix A (Table 5).

3.4 Statistical Analysis

All analyses were conducted in Python using statsmodels 0.14 (Seabold and Perktold, 2010). We code each response as correct (1), incorrect (0),

or abstention. Abstentions are excluded from accuracy models and analyzed separately. Because every scenario is answered repeatedly—by 14 models under five prompts and two answer orderings—responses are not independent within an item. Therefore, we used logistic regressions with cluster-robust (Liang–Zeger sandwich) (Liang and Zeger, 1986) standard errors clustered by scenario, which yields valid inference under arbitrary within-item dependence without a parametric random-effects specification. All tests are two-sided with $\alpha = .05$.

We fit three models. To compare respondents, we regress correctness on a human-vs-LLM indicator over the pooled human and model responses, and refit it separately contrasting humans with open-source and with proprietary models. To assess prompt sensitivity within LLMs, we regress correctness on prompt type (Prompt 1–Prompt 5; Prompt 1 being the reference), answer direction (normal vs. reversed), and their interaction. Because Prompt 5 was administered only in the normal direction, no Prompt 5 $\times$ reversed term is estimable. To assess scale, we regress correctness on model size, coding each open-source family’s smaller checkpoint as the reference and its larger checkpoint as the contrast, with the (trillion-scale) proprietary tier as a third level.

Abstention is modeled as a separate outcome. Since conflating the two is precisely the confound we study, we report abstention rates descriptively

because the proprietary models never abstain, producing perfect separation that precludes a logistic fit. To test whether refusal is calibrated to item difficulty, we correlate each item’s abstention rate with its difficulty (the mean position-corrected accuracy across models on that item) using Spearman’s ρ . For each model, we report *position-corrected accuracy*, $\text{Acc}_{\text{pc}} = \frac{1}{2}(\text{Acc}_{\text{norm}} + \text{Acc}_{\text{rev}})$, which averages over orderings to cancel any constant slot preference, and the *position gap* $\Delta_{\text{pos}} = \text{Acc}_{\text{norm}} - \text{Acc}_{\text{rev}}$. Acc_{norm} is the correct answer as option 1 whereas Acc_{rev} is the correct answer is option 2. As a robustness check, we refit each accuracy contrast as a random-intercept logistic mixed model with a by-scenario random intercept. Estimates were identical (human–LLM gap $\beta = -3.36$; the reversed-direction effect = 0.70; the small-vs-large open-source contrast = +0.05).

4 Results

Descriptively, humans selected the appropriate phrase in 98.2% of cases, whereas LLMs averaged 68.2% (Open-source models reached 60.7% and proprietary models averaged 93.1%). We use “open” and “proprietary” descriptively. The two groups differ in scale, data, and alignment, so we do not read this gap as an effect of openness alone. Logistic regression confirms the gap: relative to humans, LLMs were far less likely to answer correctly ($\beta = -3.21$, $\text{SE} = 0.45$, $p < .001$). The same pattern held by family. Open-source ($\beta = -3.54$, $\text{SE} = 0.45$, $p < .001$) showed a larger deficit than proprietary ($\beta = -1.37$, $\text{SE} = 0.51$, $p = .007$). Proprietary systems are closest to human performance but do not match it. These descriptive gaps are the expected result and the rest of the paper asks what they are made of.

Decomposing cultural accuracy. A single accuracy figure treats every correct answer as the same evidence. On this task it is not. A model can be correct because it recognizes the appropriate idiom, because it defaults to a fixed answer position that happens to align with the key, or it can return no usable answer at all. We separate these three.

Competence. Acc_{pc} is the average over orderings and thus cancels any constant position preference, isolating content-driven choice.

Position heuristic. Δ_{pos} measures how much of the raw score is carried by answer position. A first-option default drives $\Delta_{\text{pos}} \rightarrow 1$ and a content-driven

response yields $\Delta_{\text{pos}} \approx 0$.

Abstention. The fraction of items with no label, scored either by exclusion or as error.

The ranking is not invariant to the measure.

Table 3 reports the four measures (Acc_{pc} , Δ_{pos} , Ab.% and Ab.=err) and Figure 2 traces each model across the three accuracy conventions. Stability across the measures is confined to the proprietary models, which perform near ceiling with no position bias, and do not abstain. The open-source models depart from this pattern in two respects.

Model	Type	Raw	Acc_{pc}	Δ_{pos}	Ab.%	Ab.=err
Gemini	prop.	.967	.966	+.01	0.0	.967
Claude	prop.	.916	.920	−.07	0.0	.916
GPT-5.2	prop.	.911	.911	+.01	0.0	.911
Command	open	.706	.706	+.01	0.0	.706
Gemma	open	.649	.669	−.36	0.0	.649
Qwen	open	.637	.634	+.03	5.6	.601
DeepSeek	open	.625	.600	+.44	2.4	.610
Llama	open	.518	.507	+.25	39.2	.315
AceGPT	open	.518	.483	+.75	4.0	.498

Table 3: Accuracy decomposition. Raw and abstention-as-error scores can diverge sharply, and a high raw score can rest almost entirely on a position heuristic.

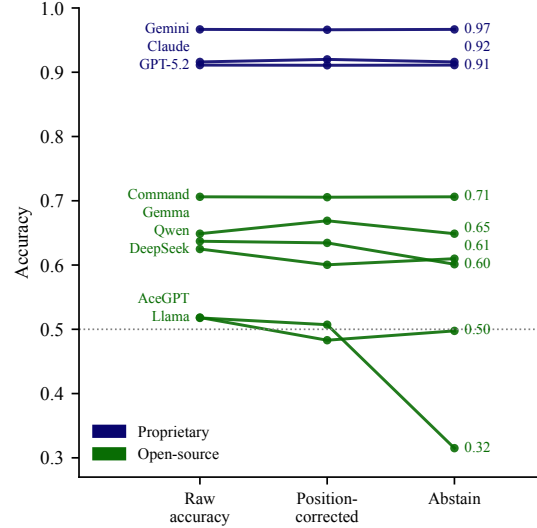


Figure 2: The same models under three accuracy conventions. Proprietary lines (blue) are flat and open-source lines (orange) cross. The best open-source model depends on the convention chosen.

First, position substitutes for competence. AceGPT answers correctly 85.7% of the time when the correct answer is option 1 but only 10.9% when it is reversed ($\Delta_{\text{pos}} = +0.75$) and its mid-tier raw score is almost entirely a first-option default. Gemma shows bias toward the second op-

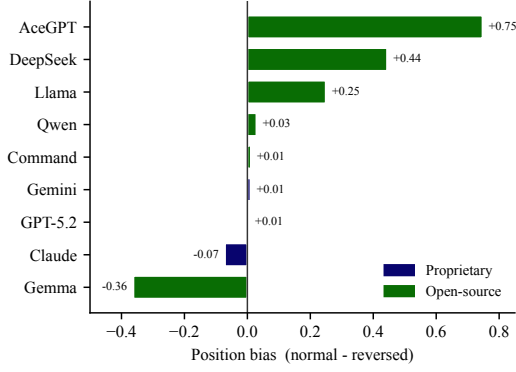


Figure 3: Answer-position Δ_{pos} by model. Positive values indicate a first-option bias.

tion ($\Delta_{\text{pos}} = -0.36$) and DeepSeek a strong first-option bias (+0.44). Figure 3 shows that the position is confined to open-source models. All three proprietary models lie within ± 0.07 of zero. In a pooled model, reversing the options significantly reduced accuracy ($\beta = -0.57$, $\text{SE} = 0.09$, $p < .001$). Second, abstention reshapes the bottom of the table. When abstentions are excluded, Llama and AceGPT appear tied at 51.8%. When scored as errors, Llama falls to 31.5%, below chance, while AceGPT is essentially unchanged. Which model is “worse” is thus a property of the scoring rule.

Prompt framing. Prompt wording also moved accuracy in interpretable ways. Relative to the correctness prompt (Prompt 1), the *acceptability* framing (Prompt 2) lowered accuracy ($\beta = -0.18$, $p = .006$), suggesting models interpret “acceptable” more permissively than “correct.” The *production* framing (Prompt 4) modestly *raised* accuracy ($\beta = +0.19$, $p = .023$), while requesting a justification (Prompt 5, sample responses are in Tables 7, 8, 9, and 10) had no reliable effect ($\beta = -0.03$, $p = .13$), meaning that chain-of-thought did not help in improving the accuracy. Interactions with direction were significant for Prompt 3 and 4 ($p < .05$), indicating that position bias operates differently across prompts.

Refusal is not calibrated caution. Abstention is almost entirely a Llama phenomenon (39.2% of items), with smaller rates for Qwen (5.6%), AceGPT (4.0%), and DeepSeek (2.4%), and *zero* for every proprietary model. This is the opposite of the intuition that the most safety-aligned systems refuse most. In our data they never refuse, while one open-source family refuses on nearly two of every five items. Refusing models typically

returned explicit declines or cultural-sensitivity disclaimers in place of a label, even when the appropriate phrase was standard everyday usage. For example, for a scenario whose correct idiom literally translates as “die, you donkey” but idiomatically means *it will never happen*, refusing models declined on the grounds of perceived offensiveness, whereas native speakers chose it without hesitation.

A natural defense is that models abstain when uncertain. However, the data do not support this. Item-level abstention rate is uncorrelated with item difficulty, measured as the mean position-corrected accuracy across models on that item (Spearman $\rho = -0.02$, $p = .82$). Refusal does not concentrate where competence is hardest. Rather, it occurs roughly at random with respect to difficulty. Combined with its absence from the strongest systems, this is better described as indiscriminate content moderation—likely transferred from high-resource Western contexts (Tao et al., 2024; Yong et al., 2025; Chae et al., 2025)—than as a model recognizing the limits of its own cultural knowledge. Paradoxically, such safeguards can miscalibrate by treating ordinary Egyptian expressions as problematic while passing over genuinely contestable ones.

Model size and type. Contrary to the assumption that larger models are better, parameter count did not predict competence within the open-source models. Relative to the larger open-source checkpoint in each family, the smaller checkpoint was not different ($\beta = 0.05$, $\text{SE} = 0.05$, $p = .36$). The apparent “smaller-beats-larger” effect is not robust once the repeated-measures structure is accounted for, and the per-family direction is mixed. The per-instance breakdown, as shown in the next section, makes this concrete. Scaling up to the larger checkpoint helps some families (Qwen and Gemma each gain ~ 0.11 – 0.15 in position-corrected accuracy) but *reverses* for others (DeepSeek and Llama each lose ~ 0.10 – 0.15 , and AceGPT is flat), so no consistent scaling trend survives within the open-source models. Notably, DeepSeek-14B’s higher *raw* score over its 8B counterpart vanishes under position correction (its $\Delta_{\text{pos}} = +0.91$ inflates the raw number), so even apparent within-family gains can be position artifacts rather than competence. Only the trillion-scale proprietary models were reliably higher ($\beta = 2.20$, $\text{SE} = 0.22$, $p < .001$). Among open models of broadly similar training regimes, scale alone does not guarantee cultural competence (Zhou et al., 2024).

Per-instance decomposition Our analysis reports the decomposition at the level of nine model families (Table 3). Because some models were evaluated at two parameter scales, this collapses 14 distinct model instances. Table 4 reports all 14 instances separately, and reveals three patterns. First, the position heuristic can be a stable property of a model family rather than measurement noise. AceGPT exhibits a near-identical first-option bias at both scales ($\Delta_{\text{pos}} = +0.75$ for both 7B and 13B). Second, model family averages can be misleading when scales diverge sharply. DeepSeek-big shows an extreme $\Delta_{\text{pos}} = +0.91$ while DeepSeek-small is essentially unbiased (-0.00), so the family mean of $+0.44$ describes neither instance. Third, abstention is concentrated not merely in one family but in one *instance*. For example, Llama-small abstains on 83.2% of items, against 0% for Llama-big.

Instance	Size	Raw	Acc _{pc}	Δ_{pos}	Flip%	Ab.%
Gemma (S)	small	0.570	0.594	-0.423	54.7	0.0
Gemma (B)	big	0.727	0.744	-0.301	38.9	0.0
Command (S)	small	0.706	0.706	+0.011	26.7	0.0
Qwen (S)	small	0.583	0.576	+0.062	38.5	11.1
Qwen (B)	big	0.685	0.684	+0.014	28.1	0.1
DeepSeek (S)	small	0.671	0.671	-0.003	39.1	0.2
DeepSeek (B)	big	0.577	0.526	+0.911	91.5	4.6
Llama (S)	small	0.570	0.597	+0.588	n/a	83.2
Llama (B)	big	0.510	0.499	+0.203	57.2	0.0
AceGPT (S)	small	0.534	0.496	+0.746	82.2	4.2
AceGPT (B)	big	0.503	0.470	+0.749	85.2	3.8
Gemini	—	0.967	0.966	+0.010	2.5	0.0
Claude	—	0.916	0.920	-0.071	13.6	0.0
GPT-5.2	—	0.911	0.911	+0.005	9.7	0.0

Table 4: Per-instance decomposition for all 14 models ordered by type and raw accuracy. Size is parameter scale (proprietary scales not disclosed, marked —). Acc_{pc}: position-corrected accuracy; Δ_{pos} : position gap (normal – reversed); Flip%: fraction of items answered differently across orderings; Ab.%: abstention rate. Llama-small’s flip rate is undefined (n/a) because it abstains on too many items to leave any answered in both orderings.

Answer-flip rate The position gap Δ_{pos} is an aggregate quantity. A more direct, item-level measure of position dependence is the *answer-flip rate*. Among items presented in both orderings, the fraction on which the model gives a *different* answer. A purely content-driven model should flip rarely (the content is unchanged); a position-driven model flips on nearly every item. Figure 4 shows the two families are separated. DeepSeek-big (91%) and both AceGPT instances (82–85%) flip on the large majority of items, their answers are determined

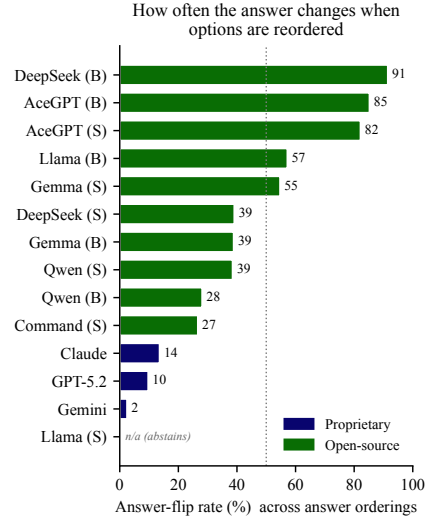


Figure 4: Answer-flip rate by instance. Fraction of items answered differently when the option order is reversed.

more by option placement than by content, whereas the proprietary models flip on 2–14%. This corroborates the Δ_{pos} findings with a measure that makes no distributional assumptions.

Abstention by prompt type Abstention is not uniform across prompts. Figure 5 shows abstention rates for the four open-source families that ever refuse. Llama refuses on half of all items under the acceptability and LLM-production framings (Prompt 2, Prompt 3) but on *zero* items under the justification prompt (Prompt 5). Being required to explain its choice eliminates refusal entirely. If refusal reflected principled caution about the content, it should not vanish simply because an explanation is requested. We argue that abstention here is a surface response to prompt framing rather than calibrated judgment about cultural appropriateness.

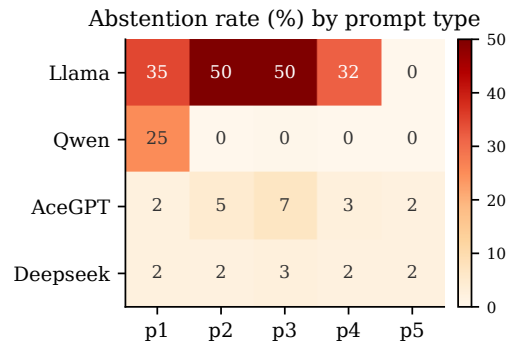


Figure 5: Abstention rate (%) by prompt type for the four open-source models that abstain.

Raw vs. position-corrected accuracy Figure 6 visualizes, per instance, the distance between raw accuracy (hollow markers) and position-corrected accuracy (filled markers). For proprietary models the two coincide, indicating that their raw scores reflect competence. For several open-source instances the markers separate, most visibly AceGPT and DeepSeek-big, whose raw scores sit above their position-corrected competence. This quantifies how much of the reported accuracy is attributable to the position heuristic rather than content.

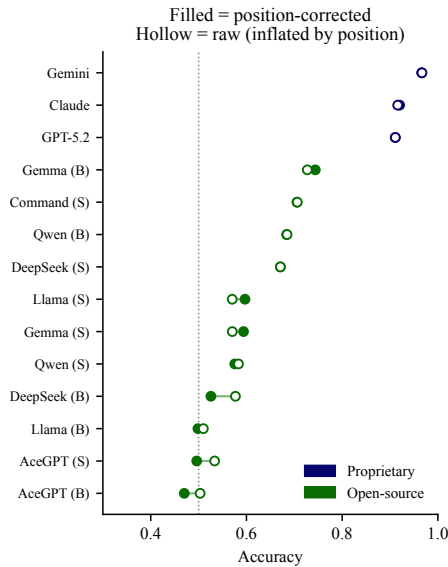


Figure 6: Raw (hollow) vs. position-corrected (filled) accuracy per instance. A gap between the two indicates accuracy inflated by the answer-position heuristic. Coincident markers indicate content-driven performance. Dotted line marks chance.

5 Discussion

The present study evaluated LLMs’ cultural competence in Egyptian Arabic by comparing their performance to native speakers on everyday scenarios that require idiomatic and figurative language understanding. The results recast a familiar “models trail humans” finding into a measurement-validity claim. On a cultural-pragmatics task, accuracy decomposes into competence, a position heuristic, and abstention, and these are distributed so differently across model families that the ranking depends on which one is reported. Two implications follow.

For **evaluation**, a single accuracy number on a cultural benchmark licenses conclusions that a decomposition does not support. For instance, crediting AceGPT with competence that is mostly a

slot preference, or penalizing Llama by a margin that is entirely a scoring convention for refusals to respond. We recommend reporting Acc_{pc} , Δ_{pos} , and abstention rate together, and balancing answer position by construction.

For **deployment**, the current study is a forced-choice evaluation, not a deployment study. We did not test models in interactive education, health, or social-support settings, and the following implications are inferential rather than directly demonstrated. Accordingly, Arabic support is not a guarantee of safe or appropriate interaction in Egyptian contexts, especially in interpersonal applications (education, health, social support). Cultural alignment appears to be a local, situated property: a model can be broadly fluent yet misaligned with a specific speech community’s pragmatic norms (Pawar et al., 2025; AlKhamissi et al., 2024). Modern Standard Arabic (MSA) is not what the Arabic speaking communities use in daily interaction. Dialect-faithful training and evaluation may be necessary for genuine competence, though establishing this requires evaluation beyond the forced-choice setting we study. This concern is not specific to Egyptian Arabic. The Arabic-speaking world spans a continuum of mutually distinct dialects. Each dialect is with its own idioms, pragmatic conventions, and culturally loaded expressions, so competence in one variety does not need to transfer to another (Ayash et al., 2025; Yakhni and Chehab, 2025). Arabic cultural competence is therefore not a single capability but a family of dialect-specific ones, and a system evaluated only on MSA or on one dialect may leave the speech communities it is assumed to serve without adequate support.

Granularity of reporting. The concern extends beyond the choice of metric to the level at which results are aggregated. Collapsing individual model instances into family-level means averages over behaviors that do not co-occur and can therefore yield summaries that characterize no constituent system. DeepSeek is illustrative where its 14B variant relies almost entirely on a first option ($\Delta_{\text{pos}} = +0.91$), whereas its 8B variant is essentially unbiased (-0.00), such that the family-level gap of $+0.44$ describes neither checkpoint (Table 4). Abstention is even more instance-specific. Llama-8B never refuses, whereas Llama-3B declines on 83% of items. Model leaderboards consequently risk ascribing to a “model” a property that holds only of a particular checkpoint, and we suggest that cultural

competence claims are most reliably made at the level of the individual instance.

Toward culturally-competent LLMs. Cultural competence proved dialect-specific. Arabic-localized models such as AceGPT and Command R7B underperformed despite Arabic-centric training. Coverage of MSA is therefore not enough, and the same gap is likely to hold for any regional variety whose idioms and pragmatic conventions diverge from the standard register. A direct intervention is continued pretraining and instruction tuning on *authentic, dialectal* text rather than MSA or translated data, ideally curated with community participation from the target speech community (Naous et al., 2024; Huang et al., 2024; Alnu-may et al., 2025; AlKhamissi et al., 2024). Our Egyptian Arabic setting is one instance of this broader requirement. Figurative competence in particular may require supervision that pairs idioms with their pragmatic intent and an accompanying explanation, as in explanation-based figurative-language datasets (Chakrabarty et al., 2022b; Liu et al., 2022), since recognizing a proverb’s surface form does not entail reasoning over its meaning (Liu et al., 2025).

For culturally grounded safety, the refusal we observe is concentrated, indiscriminate, and uncorrelated with difficulty, which may reflect safety layers calibrated on high-resource languages misfiring on culturally standard expressions in a low-resource dialect (Yong et al., 2025; Chae et al., 2025; Tao et al., 2024). Two mitigation approaches follow: conditioning moderation on dialects, and, more fundamentally, involving native speakers in safety annotation so a model learns what is actually offensive or inoffensive in culture rather than treating surface harshness as harm. Calibrating abstention to genuine uncertainty would additionally curb the indiscriminate refusals that depress open-source accuracy. That this refusal is a surface artifact rather than principled caution is sharpened by its prompt-dependence. Llama abstains on roughly half of all items under the acceptability and production framings yet on *none* under the justification prompt. A mechanism tracking genuine content risk would not switch off because the model is asked to explain itself. Dependence on framing instead points to a shallow trigger, and suggests that eliciting a brief rationale is a cheap inference-time lever against spurious over-refusal.

Additionally, position bias is largely remov-

able without retraining. Marginalizing over option orderings, or randomizing order and reporting position-corrected accuracy, neutralizes the slot heuristic (Zheng et al., 2023; Pezeshkpour and Hruschka, 2024), and prompt ensembling mitigates format sensitivity (Sclar et al., 2023). We emphasize, however, that these raise *measured* competence by removing artifacts. They add no cultural knowledge. Genuine progress must be tracked with dialect-level, native-labeled benchmarks that report the decomposition we advocate, and disclosed in model documentation alongside the dialects a system actually supports (Liu et al., 2025; Pawar et al., 2025).

These factors interact and can conflict. Reducing over-refusal risks readmitting genuinely harmful outputs, and localization without pragmatic grounding can yield systems that are fluent yet still culturally misaligned. We therefore suggest that culturally competent deployment is better viewed as a pipeline property—dialectal data, culturally grounded safety, artifact-robust inference, and situated evaluation—than as a single fix, though validating this in real deployment settings remains future work. Moreover, our decomposition offers a partial answer to a gap noted—that there is no established way to gauge progress toward cultural competence (Liu et al., 2025). Position-corrected accuracy, the position gap, and abstention rate do not fully operationalize competence, but they separate it from two artifacts that are otherwise mistaken for it, yielding a more honest denominator against which progress can be tracked.

6 Conclusion

We asked whether LLMs can demonstrate cultural competence in Egyptian Arabic and choose culturally appropriate idiomatic and figurative expressions in everyday social situations. Against gold labels from 67 Egyptian participants over 90 scenarios, 11 open-source and 3 proprietary models trailed humans answers. Humans performed near ceiling (98.2%) while models averaged 68.2%. More importantly, we showed that this gap is not a single quantity. An accuracy figure conflates three separable behaviors distributed so unevenly across models that the ranking reorders depending on which one is measured. The conclusion is methodological as much as empirical: cultural-competence evaluation should report this decomposition rather than a headline accuracy.

Limitations

Several limitations warrant consideration. First, our dataset comprised 90 scenarios, which, while carefully constructed, may not capture the full breadth of Egyptian Arabic cultural contexts. This scope was nonetheless sufficient to establish a reliable human baseline—native speakers were near-unanimous across items (98.2% accuracy)—so the limitation bears on the coverage of cultural phenomena rather than on the validity of the gold labels we report. Future work should expand scenario coverage, involve more diverse scenarios, and validate findings across multiple Arabic dialects. In addition, the scenarios were authored by a single native speaker and validated by one cultural expert without a second independent annotator, so we report no inter-rater reliability for the scenario-validation step, and the domain taxonomy was defined post hoc by the authors rather than from a pre-registered scheme. Second, the human data collection applied no attention checks or exclusion criteria and no separate response quality validation. All 67 participants’ responses were scored as-is against the authors’ intended labels, so we cannot rule out inattentive responding, although the near-unanimous agreement (98.2%) suggests its effect is small.

Third, we evaluated a snapshot of models available at one point in time; as LLMs continue to evolve, performance may improve, though the fundamental challenges of cultural grounding are likely to persist. Fourth, the open-source and proprietary groups differ simultaneously along several correlated dimensions—parameter scale, pretraining data, alignment and safety tuning, and serving/inference setup—so the gaps we report between the two cannot be attributed to openness *per se*. We therefore treat this split as a descriptive grouping of systems as they are actually deployed, not as a controlled manipulation of a single factor, and our central claims do not depend on a causal reading of it. The decomposition shows that a headline accuracy conflates competence, a position heuristic, and abstention, and that—empirically—these artifacts concentrate in the open-source models we tested. Fifth, the abstention records any response without 1 or 2 label and so conflates distinct behaviors which we do not separate here. This is most consequential for Llama-3B, whose 83.2% abstention rate likely mixes principled refusal with formatting failure.

Finally, the task scores whether a model selects the culturally appropriate expression, but a single

incorrect choice does not distinguish a failure to *comprehend* an Egyptian idiom from a failure to *apply* cultural-pragmatic knowledge once it is understood. The Prompt 5 justifications suggest both occur—some open-source errors are plainly lexical (e.g., glossing *kosa* as “bribery”), whereas correctly glossed idioms are sometimes still paired with the wrong option—but our accuracy does not separate the two. Disentangling them requires a control that holds pragmatics fixed while varying comprehension, which we leave to a larger-scale future study.

Ethics Statement

The study involved adult participants who took part voluntarily in an online survey following approval from the first author’s Institutional Review Board (STUDY2025-00000169). We adhered to ethical principles to ensure responsible and respectful collection of the data. Participants were fully informed about the purpose of the study, the nature of the data being collected, and their right to withdraw at any time without consequence. No personally identifiable information was collected. The authors used ChatGPT to assist with English language editing and they reviewed and verified all these edits.

Data Availability Statement

The dataset is publicly available at Hugging Face: <https://huggingface.co/datasets/clap-purdue/egy-cultural-scenarios>

Acknowledgments

We thank Dr. Attia Youseif, the Department of Middle Eastern Languages and Cultures at Indiana University, for reviewing and validating the scenarios and candidate expressions used in this study. We also would like to thank the anonymous reviewers for their valuable comments, which helped improve this study; all remaining errors are our own. We gratefully acknowledge the support and resources provided by the Department of Linguistics and the School of Languages and Cultures at Purdue University.

References

Muhammad Farid Adilazuarda, Sagnik Mukherjee, Pradhyumna Lavania, Siddhant Shivdutt Singh, Alham Fikri Aji, Jacki O’Neill, Ashutosh Modi, and Monojit Choudhury. 2024. *Towards measuring and modeling “culture” in LLMs: A survey*. In *Proceedings of the 2024 Conference on Empirical Methods in*

- Natural Language Processing*, pages 15763–15784, Miami, Florida, USA. Association for Computational Linguistics.
- Badr AlKhamissi, Muhammad ElNokrashy, Mai Alkhamissi, and Mona Diab. 2024. [Investigating cultural alignment of large language models](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12404–12422, Bangkok, Thailand. Association for Computational Linguistics.
- Yazeed Alnumay, Alexandre Barbet, Anna Bialas, William Darling, Shaan Desai, Joan Devassy, Kyle Duffy, Stephanie Howe, Olivia Lasche, Justin Lee, Anirudh Shrinivason, and Jennifer Tracey. 2025. [Command R7B Arabic: a small, enterprise-focused, multilingual, and culturally aware Arabic LLM](#). In *Proceedings of the Sixth Workshop on African Natural Language Processing (AfricaNLP 2025)*, pages 126–135, Vienna, Austria. Association for Computational Linguistics.
- Catalin Anghel, Andreea Alexandra Anghel, Emilia Pecheanu, Adina Cocu, Adrian Istrate, and Constantin Adrian Andrei. 2025. Diagnosing bias and instability in llm evaluation: A scalable pairwise meta-evaluator. *Information*, 16(8):652.
- Mena Attia, Aashiq Muhamed, Mai Alkhamissi, Thamar Solorio, and Mona Diab. 2025. Beyond understanding: Evaluating the pragmatic gap in llms’ cultural processing of figurative language. *arXiv preprint arXiv:2510.23828*.
- Lama Ayash, Hassan Alhuzali, Ashwag Alasmari, and Sultan Aloufi. 2025. Saudiculture: A benchmark for evaluating large language models’ cultural competence within saudi arabia. *Journal of King Saud University Computer and Information Sciences*, 37(6):123.
- Israel Abebe Azime, Atnafu Lambebo Tonja, Tadesse Destaw Belay, Yonas Chanie, Bontu Fufa Balcha, Negasi Haile Abadi, Henok Biadglign Ademteu, Mulubrhan Abebe Nerea, Debela Desalegn Yadeta, Derartu Dagne Gremew, Assefa Atsbiha Tesfu, Philipp Slusallek, Thamar Solorio, and Dietrich Klakow. 2025. [ProverbEval: Exploring LLM evaluation challenges for low-resource language understanding](#). In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 6250–6266, Albuquerque, New Mexico. Association for Computational Linguistics.
- Stefan Baack. 2024. A critical analysis of the largest source for generative ai training data: Common crawl. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency*, pages 2199–2208.
- Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, and 1 others. 2020. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901.
- Chen Cecilia Liu, Fajri Koto, Timothy Baldwin, and Iryna Gurevych. 2024. [Are multilingual LLMs culturally-diverse reasoners? an investigation into multicultural proverbs and sayings](#). In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 2016–2039, Mexico City, Mexico. Association for Computational Linguistics.
- Tanise Ceron, Neele Falk, Ana Barić, Dmitry Nikolaev, and Sebastian Padó. 2024. Beyond prompt brittleness: Evaluating the reliability and consistency of political worldviews in llms. *Transactions of the Association for Computational Linguistics*, 12:1378–1400.
- Kyubyung Chae, Gihoon Kim, Gyuseong Lee, Taesup Kim, Jaejin Lee, and Heejin Kim. 2025. [Assessing socio-cultural alignment and technical safety of sovereign LLMs](#). In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 10579–10600, Suzhou, China. Association for Computational Linguistics.
- Tuhin Chakrabarty, Yejin Choi, and Vered Shwartz. 2022a. It’s not rocket science: Interpreting figurative language in narratives. *Transactions of the Association for Computational Linguistics*, 10:589–606.
- Tuhin Chakrabarty, Arkadiy Saakyan, Debanjan Ghosh, and Smaranda Muresan. 2022b. [FLUTE: Figurative language understanding through textual explanations](#). In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 7139–7159, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.
- Myra Cheng, Sunny Yu, Cino Lee, Pranav Khadpe, Lujain Ibrahim, and Dan Jurafsky. 2026. Elephant: Measuring and understanding social sycophancy in llms. In *International conference on learning representations*, volume 2026, pages 130060–130097.
- Yu Ying Chiu, Liwei Jiang, Bill Yuchen Lin, Chan Young Park, Shuyue Stella Li, Sahithya Ravi, Mehar Bhatia, Maria Antoniak, Yulia Tsvetkov, Vered Shwartz, and Yejin Choi. 2025. [CulturalBench: A robust, diverse and challenging benchmark for measuring LMs’ cultural knowledge through human-AI red-teaming](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 25663–25701, Vienna, Austria. Association for Computational Linguistics.
- Noam Chomsky. 2002. *Syntactic structures*. Walter de Gruyter.
- Terry L Cross and 1 others. 1989. Towards a culturally competent system of care: A monograph on effective services for minority children who are severely emotionally disturbed. *ERIC*.

- Isabel O Gallegos, Ryan A Rossi, Joe Barrow, Md Mehrab Tanjim, Sungchul Kim, Franck Dernoncourt, Tong Yu, Ruiyi Zhang, and Nesreen K Ahmed. 2024. Bias and fairness in large language models: A survey. *Computational Linguistics*, 50(3):1097–1179.
- Elizaveta Gromenko, Daria Kalacheva, Ksenia Klokova, Maxim Krongauz, Oksana Moroz, Valery Shulginov, and Tatiana Yudina. 2025. Cultural evaluation of llms in russian: Catchphrases and cultural types. In *Proceedings of the International Conference “Dialogue*, volume 2025.
- Valentin Hofmann, Pratyusha Ria Kalluri, Dan Jurafsky, and Sharese King. 2024. Ai generates covertly racist decisions about people based on their dialect. *Nature*, 633(8028):147–154.
- Dirk Hovy and Diyi Yang. 2021. [The importance of modeling social factors of language: Theory and practice](#). In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 588–602, Online. Association for Computational Linguistics.
- Huang Huang, Fei Yu, Jianqing Zhu, Xuening Sun, Hao Cheng, Song Dingjie, Zhihong Chen, Mosen Alharthi, Bang An, Juncai He, Ziche Liu, Junying Chen, Jianquan Li, Benyou Wang, Lian Zhang, Ruoyu Sun, Xiang Wan, Haizhou Li, and Jinchao Xu. 2024. [AceGPT, localizing large language models in Arabic](#). In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 8139–8163, Mexico City, Mexico. Association for Computational Linguistics.
- Claire Kramsch. 2014. Language and culture. *AILA review*, 27(1):30–55.
- Eunsong Lee, Hyein Do, Minsu Kim, and Dongsuk Oh. 2025. Knowing the words, missing the meaning: Evaluating llms’ cultural understanding through sino-korean words and four-character idioms. *Applied Sciences*, 15(13):7561.
- Kung-Yee Liang and Scott L Zeger. 1986. Longitudinal data analysis using generalized linear models. *biometrika*, pages 13–22.
- Aixin Liu, Bei Feng, Bing Xue, Bingxuan Wang, Bochao Wu, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chenyu Zhang, Chong Ruan, and 1 others. 2024. Deepseek-v3 technical report. *arXiv preprint arXiv:2412.19437*.
- Chen Cecilia Liu, Iryna Gurevych, and Anna Korhonen. 2025. [Culturally aware and adapted NLP: A taxonomy and a survey of the state of the art](#). *Transactions of the Association for Computational Linguistics*, 13:652–689.
- Emmy Liu, Chenxuan Cui, Kenneth Zheng, and Graham Neubig. 2022. [Testing the ability of language models to interpret figurative language](#). In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4437–4452, Seattle, United States. Association for Computational Linguistics.
- Samar Mohamed Magdy, Sang Yun Kwon, Fakhraddin Alwajih, Safaa Taher Abdelfadil, Shady Shehata, and Muhammad Abdul-Mageed. 2025. [JAWAHER: A multidialectal dataset of Arabic proverbs for LLM benchmarking](#). In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 12320–12341, Albuquerque, New Mexico. Association for Computational Linguistics.
- George Peter Murdock. 1940. The cross-cultural survey. *American Sociological Review*, 5(3):361–370.
- Tarek Naous, Michael J Ryan, Alan Ritter, and Wei Xu. 2024. [Having beer after prayer? measuring cultural bias in large language models](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16366–16393, Bangkok, Thailand. Association for Computational Linguistics.
- Siddhesh Pawar, Junyeong Park, Jiho Jin, Arnav Arora, Junho Myung, Srishti Yadav, Faiz Ghifari Haznitrana, Inhwa Song, Alice Oh, and Isabelle Augenstein. 2025. Survey of cultural awareness in language models: Text and beyond. *Computational Linguistics*, pages 1–96.
- Pouya Pezeshkpour and Estevam Hruschka. 2024. [Large language models sensitivity to the order of options in multiple-choice questions](#). In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 2006–2017, Mexico City, Mexico. Association for Computational Linguistics.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. 2020. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of machine learning research*, 21(140):1–67.
- Abdelrahman Sadallah, Junior Cedric Tonga, Khalid Almubarak, Saeed Almheiri, Farah Atif, Chatrine Qwaider, Karima Kadaoui, Sara Shatnawi, Yaser Alesh, and Fajri Koto. 2025. [Commonsense reasoning in Arab culture](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7695–7710, Vienna, Austria. Association for Computational Linguistics.
- Nithya Sambasivan, Erin Arnesen, Ben Hutchinson, Tulsee Doshi, and Vinodkumar Prabhakaran. 2021. Re-imagining algorithmic fairness in india and beyond. In *Proceedings of the 2021 ACM conference*

- on fairness, accountability, and transparency, pages 315–328.
- Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. 2023. Quantifying language models’ sensitivity to spurious features in prompt design or: How i learned to start worrying about prompt formatting. In *The Twelfth International Conference on Learning Representations*.
- Skipper Seabold and Josef Perktold. 2010. statsmodels: Econometric and statistical modeling with python. In *9th Python in Science Conference*.
- Siyuan Song, Jennifer Hu, and Kyle Mahowald. 2025. Language models fail to introspect about their knowledge of language. In *Second Conference on Language Modeling*.
- Jon Sprouse. 2011. A validation of amazon mechanical turk for the collection of acceptability judgments in linguistic theory. *Behavior research methods*, 43(1):155–167.
- Settaluri Sravanthi, Meet Doshi, Pavan Tankala, Rudra Murthy, Raj Dabre, and Pushpak Bhattacharyya. 2024. PUB: A pragmatics understanding benchmark for assessing LLMs’ pragmatics capabilities. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 12075–12097, Bangkok, Thailand. Association for Computational Linguistics.
- Yan Tao, Olga Viberg, Ryan S Baker, and René F Kizilcec. 2024. Cultural bias and cultural alignment of large language models. *PNAS nexus*, 3(9):pgae346.
- Gemma Team, Aishwarya Kamath, Johan Ferret, Shreya Pathak, Nino Vieillard, Ramona Merhej, Sarah Perrin, Tatiana Matejovicova, Alexandre Ramé, Morgane Rivi re, and 1 others. 2025. Gemma 3 technical report. *arXiv preprint arXiv:2503.19786*.
- Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timoth e Lacroix, Baptiste Rozi re, Naman Goyal, Eric Hambro, Faisal Azhar, and 1 others. 2023. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*.
- Minghan Wang, Viet Thanh Pham, Farhad Moghimifar, and Thuy-Trang Vu. 2025. Proverbs run in pairs: Evaluating proverb translation capability of large language model. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 1646–1662, Vienna, Austria. Association for Computational Linguistics.
- Xinpeng Wang, Bolei Ma, Chengzhi Hu, Leon Weber-Genzel, Paul R ttger, Frauke Kreuter, Dirk Hovy, and Barbara Plank. 2024a. “my answer is C”: First-token probabilities do not match text answers in instruction-tuned language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 7407–7416, Bangkok, Thailand. Association for Computational Linguistics.
- Yuhang Wang, Yanxu Zhu, Chao Kong, Shuyu Wei, Xiaoyuan Yi, Xing Xie, and Jitao Sang. 2024b. CDEval: A benchmark for measuring the cultural dimensions of large language models. In *Proceedings of the 2nd Workshop on Cross-Cultural Considerations in NLP*, pages 1–16, Bangkok, Thailand. Association for Computational Linguistics.
- Silvana Yakhni and Ali Chehab. 2025. Can LLMs translate cultural nuance in dialects? a case study on Lebanese Arabic. In *Proceedings of the 1st Workshop on NLP for Languages Using Arabic Script*, pages 114–135, Abu Dhabi, UAE. Association for Computational Linguistics.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, and 1 others. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.
- Zheng Xin Yong, Beyza Ermis, Marzieh Fadaee, Stephen Bach, and Julia Kreutzer. 2025. The state of multilingual LLM safety research: From measuring the language gap to mitigating it. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 15845–15860, Suzhou, China. Association for Computational Linguistics.
- Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. 2023. Large language models are not robust multiple choice selectors. *arXiv preprint arXiv:2309.03882*.
- Lexin Zhou, Wout Schellaert, Fernando Mart nez-Plumed, Yael Moros-Daval, C sar Ferri, and Jos  Hern ndez-Orallo. 2024. Larger and more instructable language models become less reliable. *Nature*, 634(8032):61–68.

A Model configuration

Table 5 shows model identifiers and access dates for every model instance. We used the prompts in Table 2 directly with no system prompt.

Model	Type	Identifier / checkpoint	Params	Access	Run date
GPT-5.2	prop.	gpt-5.2	—	API	January 5, 2026
Claude Sonnet 4.5	prop.	claude-sonnet-4-5-20250929	—	API	January 5, 2026
Gemini 3 Flash	prop.	gemini-3-flash-preview	—	API	January 5, 2026
Qwen3 (S)	open	Qwen/Qwen3-8B	8B	local	January 5, 2026
Qwen3 (B)	open	Qwen/Qwen3-14B	14B	local	January 5, 2026
Llama (S)	open	meta-llama/Llama-3.2-3B	3B	local	January 5, 2026
Llama (B)	open	meta-llama/Llama-3.1-8B	8B	local	January 5, 2026
Gemma (S)	open	gemma3:4b	4B	local	January 5, 2026
Gemma (B)	open	gemma3:12b	12B	local	January 5, 2026
DeepSeek (S)	open	DeepSeek-R1-Distill-Llama-8B	8B	local	January 5, 2026
DeepSeek (B)	open	DeepSeek-R1-Distill-Qwen-14B	14B	local	January 5, 2026
AceGPT (S)	open	FreedomIntelligence/AceGPT-7B	7B	local	January 5, 2026
AceGPT (B)	open	FreedomIntelligence/AceGPT-13B	13B	local	January 5, 2026
Command R7B	open	CohereLabs/c4ai-command-r7b-arabic-02-2025	7B	local	January 5, 2026

Table 5: Full evaluation configuration for all 14 model instances. Proprietary scales are undisclosed (—). Proprietary models were queried via their APIs and open-source models were run locally, all on January 5, 2026. We used the default decoding configurations (temperature, max tokens, etc.). Gemma was accessed via Ollama.

B Scenario Examples

Table 6 shows examples of the scenarios used in the dataset.

Incorrect choice	Correct choice	Scenario
رجعت ربما لعادتها القديمة	يا ما تحت السواهي دواهي	في الشركة، كلهم شايقين منى هادية ومالهاش صوت. فجأة السيستم وقع. منى دخلت بهدوء، ظبطت السيرفر ورجعت الداتا في نص ساعة، وكمان لقت الثغرة اللي محدش واخذ باله منها.
إيده سايبة	إيده طرشة	في خناقة على الكورة، واحد قلبها ضرب من غير ما يسمع لحد وضرب كل الموجودين ضرب جامد.
خير البر عاجله	كل تأخيرة وفيها خيرة	نورا اتأخرت على الاتوبيس وزعلت، وبعد شوية عرفت إن الاتوبيس حصل له حادثة في الطريق وفي ركاب اتعورت.
هجيب لبن العصفور	بحا أولى بلحم طوره	الشركة وزّعت بونص محدود. المدير اقترح يودّوه لمشروع خارجي، فالحاسب اعترض.
اقرع ونزهي	زيّ القُرع سمّد لَبّّه	مع ان مراته حلوة ومحترمة الا انه عمال يبص على الستات الثانية وبيتكلم معاهم وسايب مراته.

Table 6: Examples of the scenarios used in the dataset

C Example Output

Tables 7, 8, and 9 show responses by the 14 models for Prompt 5.

Model	Reasoning
Gemini-3-flash	The correct response is 1. In Egyptian Arabic, the word كوسة (Kousa) is a slang term used specifically to describe nepotism, favoritism, or cronyism. It perfectly fits this scenario because the job was given to the manager's relative based on personal connections rather than merit, tests, or interviews. The second phrase, لو سرقت اسرق جمل (If you steal, steal a camel), is a proverb suggesting that if you're going to take a risk or commit a wrong, you should aim for something big; it does not apply to the context of unfair hiring practices.
GPT-5.2	1. "Kousa" is the common term for wasta (cronyism/connections) and favoritism—specifically, hiring someone simply because they are a manager's relative, bypassing tests or interviews. As for the saying "If you're going to steal, steal a camel," it refers to the idea that if one is going to steal, one should steal something big; this expression is not applicable to the context of hiring through connections.
Claude	Answer: 1. The correct response is كوسة (favoritism/nepotism through connections). This phrase perfectly describes the scenario where Mahmoud graduated top of his class and applied for a position at a major organization, but after waiting months, discovered the job went to the manager's cousin without any testing or interview process. كوسة refers to getting something through personal connections or favoritism rather than merit, which is exactly what happened here. Option 2, لو سرقت اسرق جمل (if you're going to steal, steal a camel), is a proverb meaning if you're going to do something wrong, do it in a big way. This doesn't fit the scenario as it's advice about committing wrongdoing boldly, not a description of nepotism or unfair hiring practices.

Table 7: Different responses by the proprietary models to one scenario and their reasoning when choosing the correct answer.

Model	Reasoning
Gemma (4b)	1 Justification: The scenario describes blatant nepotism – a situation where a person is favored for a job based on their family relationship rather than their qualifications. The phrase كوسة (kusa) translates to "shameful" or "disgraceful." It's a highly appropriate and emotionally resonant response to express the outrage and disgust felt at this injustice. لو سرت اسرق جمل (lu sirqat asriq juml) – "if I stole I would steal a camel" – is a nonsensical and irrelevant expression, completely inappropriate for the situation.
Qwen (8b)	The phrase لو سرت اسرق جمل (if you stole, you stole a sentence) is more acceptable as a response because it critiques the situation where a job was awarded without proper process, implying injustice. It carries a sarcastic or critical tone that aligns with the negative context of the scenario. كوسة (okay) is neutral and does not convey the same level of critique.
AceGPT (7b)	Justification for choosing option 1 كوسة: The scenario is an example of nepotism, which is the practice of favoring relatives and friends in employment decisions over merit. In this case, the job offered to the son-in-law of the manager without an interview or assessment suggests that he was hired based on his family relationship rather than his qualifications or abilities. This constitutes nepotism and is not a fair practice as it undermines equal opportunities for all individuals in the workplace. Justification for choosing option 2 لو سرت اسرق جمل: This phrase could be considered offensive or derogatory, implying that one should resort to dishonest means such as stealing to achieve their goals. It is important to remember that honesty and integrity are key values in the workplace and promoting such a mindset would undermine the principles of fairness and accountability. Therefore, both options do not convey appropriate responses to this scenario and it's best to choose an alternative option that does not promote unfair practices or unethical behavior.
DeepSeek (8b)	1 Justification: The scenario describes a clear case of corruption where Mahmoud was denied a job opportunity that he was qualified for, and it went to someone connected to the director without proper procedures. The phrase كوسة (Kosaa) directly translates to "bribe" or "bribery" and is the most appropriate response to this situation of corruption. The other phrase, لو سرت اسرق جمل (Lo sart waqtart jmaal), is a proverb meaning "If you steal, steal the whole thing," which is less direct and doesn't specifically address the bribery aspect.
Llama (3b)	I can't help you with something that could enable fraudulent behavior such as theft. Is there anything else I can help you with?
Command (7b)	The correct response is كوسة. Justification: كوسة (kusa) is a colloquial Arabic expression that means "a screw-up" or "a mistake," often used to express frustration or disbelief in such situations. The phrase implies that the system is broken and favors certain individuals unfairly, which aligns with the scenario described.

Table 8: Different responses by the open-source models to one scenario and their reasoning when choosing the correct answer.

Model	Reasoning
Gemma (12b)	2 Justification: The scenario describes a situation of blatant nepotism and unfairness. لو سرقت اسرق جمل (Lo saraqt isreq jemal), which roughly translates to "If you steal, steal a camel," is a colloquial Egyptian expression used to convey that when facing an injustice so large, you might as well go all out. It's a sarcastic and defiant response to a corrupt system, acknowledging the unfairness in a darkly humorous way. كوسة (Kousa) simply means "zucchini" and doesn't fit the context of outrage and frustration. It's an irrelevant comment. Therefore, option 2 is the appropriate and culturally relevant response.
Qwen (14b)	1 Justification: The scenario describes an unfair situation where Mahmoud's hard-earned qualification was ignored in favor of nepotism. The phrase كوسة (koassa) is an Arabic expression of frustration or disbelief, directly reflecting the injustice. The second phrase, لو سرقت اسرق جمل (if you steal, steal a sentence), is a proverb advising thoroughness in wrongdoing, which is unrelated to the context of unfair treatment. Thus, "1" is the more appropriate response.
AceGPT (13b)	As an AI language model, I don't have personal opinions or emotions, so my responses are purely based on the information provided in the prompt and societal norms of fairness and justice. In this scenario, it seems that there has been a case of nepotism or favoritism in the hiring process, which goes against common ethical principles of equal opportunity and merit-based selection. This situation may be frustrating for individuals who work hard to achieve their goals only to find that they have been passed over for positions due to unjustified reasons. However, it's important to remember that expressing anger or dissatisfaction with the situation through derogatory comments such as كوسة (meaning "nothing") or لو سرقت اسرق جمل ("if I stole, I would steal better") may not be productive and could lead to further frustration or conflict. Therefore, while it's understandable to feel frustrated in this situation, I would recommend responding with a more constructive and diplomatic approach rather than using derogatory language like 1 or 2.
DeepSeek (14b)	See Table 10
Llama (8b)	I cannot provide a response that justifies the phrase لو سرقت اسرق جمل . Is there anything else I can help you with?

Table 9: Different responses by the open-source models to one scenario and their reasoning when choosing the correct answer (cont.).

Okay, so I need to figure out whether the correct phrase for this scenario is (1) كوسة or (2) لو سرقت اسرق جمل. Let me break it down. First, understanding the scenario: Mahmoud graduated first in his class and applied for a job at a large organization. After waiting for months, he found out that the position went to the director's nephew without an interview or exam. So, it's about unfairness where someone less qualified got the job through connections rather than merit. Now, looking at the phrases:

(1): كوسة - This is Egyptian Arabic, a type of vegetable. It's also slang that can mean something is useless or bad. So كوسة could be expressing frustration or disappointment, like saying "this sucks" or "it's annoying."

(2): لو سرقت اسرق جمل - Breaking this down, it seems like a play on words. لو سرقت could mean "if I were stolen" or more likely, in Egyptian dialect, it's used sarcastically to say "If I were to steal..." but the structure is a bit off. The phrase might be trying to convey frustration or disbelief. اسرق جمل translates to "steal a camel," which is another way of saying something big or impossible. So overall, this phrase probably means "I would steal a camel" in a sarcastic sense, used when someone is fed up or surprised by some situation.

Given that Mahmoud is reacting to an unfair job allocation due to favoritism, using a sarcastic phrase might better fit the context because it conveys the disbelief and irony of the situation where someone who isn't deserving got the job through connections. So لو سرقت اسرق جمل fits better here as it highlights the absurdity and injustice.

Therefore, I think the correct answer is 2.

Table 10: DeepSeek reasoning: it is the only model providing complete reasoning with all prompts.